

Energy Systems — Group Assignment 2026

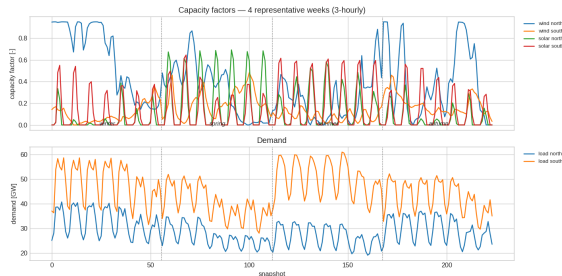
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Assignment overview

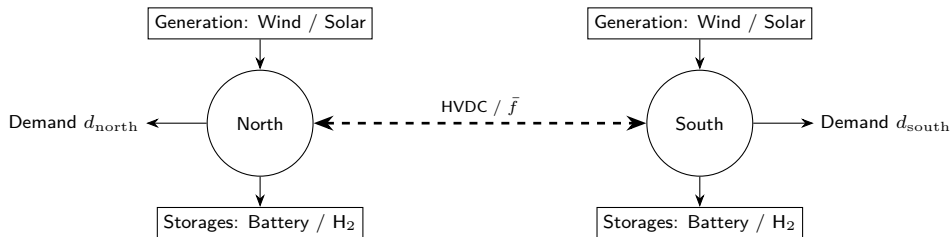
- ▶ Build a **two-node** power-system **investment + dispatch** model from scratch.
- ▶ **Two scenarios:** (i) no transmission, (ii) and transmission as a free decision variable.
- ▶ Goal: **derive insights** from the optimisation outputs — not just run code.
- ▶ Voluntary bonus project, **max 5 points**, groups of 3–5, ~2 weeks.
- ▶ **Key dates:** register by **31.05**, submit **14.06 23:59**, present **17.06 at 10:00**.

Case study: System setup and input data

- ▶ **North (DK):** wind-rich, moderate solar.
- ▶ **South (ES):** solar-rich, poor wind, larger demand.
- ▶ Both: wind, solar, H₂ (long-duration) + battery (short-duration) storage.
- ▶ Optional HVDC line (flow + = N→S).
- ▶ 224 snapshots × 3 h/snapshot (4 seasonal representative weeks).



Case study: Schematic overview



Mathematical formulation: Variables and parameters

$d_{n,t}$	demand	n	node (north / south)
$e_{n,t}^{\text{bat}}$	battery energy (state)	$p_{n,t}^{\text{bat,c}}$	battery charge power (dispatch)
$e_{n,t}^{\text{h2}}$	H ₂ energy (state)	$p_{n,t}^{\text{bat,d}}$	battery discharge power (dispatch)
$\bar{e}_n^{\text{bat}}$	battery energy capacity	$\bar{p}_n^{\text{bat}}$	battery power capacity (investment)
$\bar{e}_n^{\text{h2}}$	H ₂ energy capacity	$p_{n,t}^{\text{h2,c}}$	H ₂ charge power / electrolyser (dispatch)
f_t	line flow	$p_{n,t}^{\text{h2,d}}$	H ₂ discharge power / fuel-cell (dispatch)
$\bar{f}$	line capacity	$\bar{p}_n^{\text{el}}$	electrolyser capacity (investment)
$g_{n,t}^s$	solar dispatch	$\bar{p}_n^{\text{fc}}$	fuel-cell capacity (investment)
$g_{n,t}^w$	wind dispatch	t	snapshot index
$\text{CF}_{n,t}^s$	solar capacity factor	Δt	snapshot duration (3h)
$\text{CF}_{n,t}^w$	wind capacity factor	$\bar{g}_n^s$	solar capacity
		$\bar{g}_n^w$	wind capacity

Mathematical formulation (I): Objective

Sets: nodes $n \in \{\text{north}, \text{south}\}$, snapshots $t \in \{1, \dots, 224\}$, $\Delta t = 3 \text{ h}$.

Annuity factor: $a(r, L) = \frac{r(1+r)^L}{(1+r)^L - 1}$, $c_{\text{inv}} = \text{overnight} \cdot a(r, L) + \text{FOM}$.

Minimise system cost over the modelled window — the factor $f = T/8760 \approx 0.077$ pro-rates the annualised investment to the 672 h window:

$$\begin{aligned} \min f \bigg(\sum_n & [c_{\text{inv}}^w \bar{g}_n^w + c_{\text{inv}}^s \bar{g}_n^s + c_{\text{inv}}^{\text{el}} \bar{p}_n^{\text{el}} + c_{\text{inv}}^{\text{fc}} \bar{p}_n^{\text{fc}} + c_{\text{inv}}^{\text{h2e}} \bar{e}_n^{\text{h2}} \\ & + c_{\text{inv}}^{\text{bp}} \bar{p}_n^{\text{bat}} + c_{\text{inv}}^{\text{be}} \bar{e}_n^{\text{bat}}] + c_{\text{inv}}^{\text{line}} \bar{f} \bigg) + \Delta t \sum_{n,t} [c_{\text{var}}^w g_{n,t}^w + c_{\text{var}}^s g_{n,t}^s] \end{aligned}$$

Asymmetric H₂: *separate* electrolyser (158 186 EUR/MW/a) and fuel-cell (205 583 EUR/MW/a) costs.
Annual cost = objective/ f ; nodal price = (balance dual)/ Δt .

Mathematical formulation (II): Constraints

Power balance (north/south):
$$g_{n,t}^w + g_{n,t}^s + p_{n,t}^{h2,d} - p_{n,t}^{h2,c} + p_{n,t}^{bat,d} - p_{n,t}^{bat,c} \mp f_t = d_{n,t}$$

vRES limits:

$$g_{n,t}^w \leq CF_{n,t}^w \bar{g}_n^w, \quad g_{n,t}^s \leq CF_{n,t}^s \bar{g}_n^s$$

Transmission:

$$-\bar{f} \leq f_t \leq \bar{f}$$

H₂ (asymmetric):

$$p_{n,t}^{h2,c} \leq \bar{p}_n^{el}, \quad p_{n,t}^{h2,d} \leq \bar{p}_n^{fc}$$

Battery (symmetric):

$$p_{n,t}^{bat,c} \leq \bar{p}_n^{bat}, \quad p_{n,t}^{bat,d} \leq \bar{p}_n^{bat}$$

Energy limits:

$$e_{n,t}^{h2} \leq \bar{e}_n^{h2}, \quad e_{n,t}^{bat} \leq \bar{e}_n^{bat}$$

SOC (note Δt):

$$e_{n,t}^{h2} = e_{n,t-1}^{h2} + (\eta^{el} p_{n,t}^{h2,c} - p_{n,t}^{h2,d} / \eta^{fc}) \Delta t$$

$$e_{n,t}^{bat} = e_{n,t-1}^{bat} + (\eta^{bat} p_{n,t}^{bat,c} - p_{n,t}^{bat,d} / \eta^{bat}) \Delta t$$

Cyclic:

$$e_{n,1} = e_{n,|\mathcal{T}|}$$

Task 1 — Questions on primal solution (3 pts)

- (a) Compute the **annuity factor** $a(r, L)$ and the **annualised costs** for the technologies. Output: Table (0.5)
- (b) In the no-line case, what does the dispatch of solar and wind look like in each node? Output: Dispatch plot (0.5)
- (c) **Compare the two scenarios.** How do the investments into generation, storage, and transmission change? Output: Show the investment costs in a table and plot the capacities. (1)
- (d) How does **curtailment** by technology and region change from the no-line case to the transmission case and why? Output: Quantify and explain. (0.5)
- (e) How do the **residual load curves** change across the two scenarios? Output: Plot the residual load curves and explain the differences. (0.5)

Task 2 — Questions on dual solution (2 pts) — No plots needed

- (a) For each scenario, extract the **shadow variable to the nodal energy balances** and weight it by the nodal demand in the north and south, respectively. What do they represent? Why are they different in the two regions? When do you observe the highest values and why? (0.5)
- (b) In the transmission scenario, **consumers of which region** benefit more from the transmission line: the north, the south, or both equally? Briefly explain. (0.5)
- (c) Extract the **shadow price of the line capacity constraint** in the transmission scenario. What does it represent? In the scenario where the transmission capacity is zero, you can also extract the dual variable. Comparing the two, what does it say about the value of interconnection? (0.5)
- (d) Which advantages and disadvantages of heavy reliance on **cross-regional interconnection** does this stylised model *not* capture? No calculation needed. (0.5)

Note: prices are realistic EUR/MWh because investment is pro-rated to the window (CAPEX & OPEX on one basis); a nodal price is the balance dual $\div \Delta t$ (snapshot weighting).

Grading, submission, timeline, tools

- ▶ Introduction on **27.05.2026**.
- ▶ Work in groups of **3–5**; register via the ISIS survey by **31.05.2026, 23:59** (you may change your group until the end of the survey period).
- ▶ **5 points** (Task 1 = 3, Task 2 = 2), added to exam points; 5 pts translate into one grade step improvement (e.g. from 1.7 to 1.3).
- ▶ Report in PDF format **no longer than 8 pages** (excluding title pages and references) + reproducible notebook; final **A vs. B** comparison table. Submit via ISIS by **14.06.2026, 23:59** (late submissions are not graded).
- ▶ Presentation **17.06 at 10:00** is **mandatory** for the bonus; the group decides who presents, the others do not need to attend.
- ▶ Recommended tools: VS Code, pixi virtual environments, linopy, LaTeX for the report.
- ▶ Please do not use PyPSA (or any other integrated energy system toolboxes) to build the model, the point is to build it from scratch and understand the formulation.